

Saddle-Point Approximation and Functional Determinants

Perimeter Scholars International, Quantum Field Theory II, Tutorial 3; 15 worked parts.

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Quantum Field Theory

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Question prompts paraphrased from the official Perimeter PSI Online tutorial sheet; the equations and notation follow the source.

1. Saddle-Point Approximation

Approximation methods for ordinary integrals are among the reasons that path integrals are useful. For example, consider

$$I(z) = \int dx e^{if(x,z)}. \quad (1.1)$$

The corresponding ideas extend to functional integrals such as

$$Z[J] = \int \mathcal{D}\phi e^{iS[\phi,J]}. \quad (1.2)$$

The saddle-point method is one especially useful example.

Restore $\hbar$ in the Euclidean partition function:

$$Z[J] = \int \mathcal{D}\phi e^{-\frac{1}{\hbar}S[\phi,J]}. \quad (1.3)$$

This makes it possible to organise a semiclassical expansion by treating $\hbar$ as small. In Euclidean signature, configurations ϕ_c minimising the action dominate because configurations with larger action are exponentially suppressed. For a Minkowski integral, all stationary configurations of the action must be considered rather than only its minima.

The following questions first develop this approximation for ordinary n -dimensional integrals.

a) Let $S(x^\mu; z)$ be a real function of n spacetime variables $\mu = 1, \dots, n$ and a coupling z , and define

$$I(z) = \int_{-\infty}^{\infty} d^n x e^{-\frac{1}{\hbar}S(x;z)}. \quad (1.4)$$

Expand $S(x; z)$ around its extrema x_c and show that the sum over its minima gives

$$I(z) \approx \sum_{\text{Min}[x_c]} e^{-\frac{1}{\hbar}S(x_c;z)} \left[\frac{(2\pi\hbar)^n}{\det S''(x_c)} \right]^{1/2} + \mathcal{O}(\hbar^a), \quad (1.5)$$

where

$$[S''(x_c)]_{\mu\nu} = \left. \frac{\partial^2 S}{\partial x^\mu \partial x^\nu} \right|_{x=x_c}. \quad (1.6)$$

Solution. Expanding $S(x; z)$ around its minima, we find

$$S(x; z) = S(x_c; z) + \frac{1}{2}(x - x_c)^\mu (x - x_c)^\nu [S''(x_c; z)]_{\mu\nu} + \dots \quad (1.7)$$

In the semi-classical regime, the field configurations around the minima dominate, giving

$$\begin{aligned} I(z) &\approx \sum_{\text{Min}[x_c]} \int d^n x e^{-\frac{1}{\hbar}(S(x_c; z) + \frac{1}{2}(x-x_c)^\mu (x-x_c)^\nu [S''(x_c; z)]_{\mu\nu})} \\ &= \sum_{\text{Min}[x_c]} e^{-\frac{1}{\hbar}S(x_c; z)} \left[\frac{(2\pi\hbar)^n}{\det S''(x_c)} \right]^{1/2}, \end{aligned} \quad (1.8)$$

where we have used the results from conventional Gaussian integration in the last equality.

- b) For a generic function $S(x; z)$, determine the exponent a .

Solution. The important fluctuations should be $x - x_c \sim \sqrt{\hbar}$. The next contribution comes from S'''' (S''' is odd, and hence vanishes). Therefore, $a = n/2 + 2 - 1 = n/2 + 1$.

- c) State the condition under which the approximation becomes exact.

Solution. The approximation is exact if all higher order derivative vanishes.

- d) Explain how the saddle-point expression should change if the starting point is the Minkowski partition function

$$Z = \int e^{iS/\hbar}. \quad (1.9)$$

Solution. In real time, $-S \rightarrow iS$. Besides that, we also have to add the $i\epsilon$ prescription to ensure the path integral converges. Therefore, we have

$$I(z) \approx \sum_{\text{Min}[x_c]} e^{\frac{i}{\hbar}S(x_c; z)} \left[\frac{(2\pi i\hbar)^n}{\det(S''(x_c) + i\epsilon)} \right]^{1/2} + \mathcal{O}(\hbar^a). \quad (1.10)$$

2. Functional Determinants

Functional determinants occur throughout the path-integral treatment of QFT, including the semiclassical result above. Their formal definition as a product over operator eigenvalues is generally divergent and requires regularisation. For this exercise, set those analytic subtleties aside and manipulate the operators by analogy with matrices.

Define

$$F[f] = \log \left(\det \left[-\frac{d^2}{dy^2} + f(y) \right] \right). \quad (2.1)$$

Thus F is the logarithm of the determinant of the operator

$$\mathcal{A}[f(y)] = -\frac{d^2}{dy^2} + f(y). \quad (2.2)$$

If $M(y, z)$ is the integral kernel of an operator $\mathcal{M}$, then

$$\mathcal{M}g(y) = \int M(y, z)g(z) dz \quad (2.3)$$

for every suitable function g .

The matrix-product analogy used here is

$$\sum_y M_{xy} M_{yz} \iff \int dy M(x, y) M(y, z). \quad (2.4)$$

a) Show that the integral kernel of $\mathcal{A}$ is

$$A(y, z) = \left[-\frac{d^2}{dy^2} + f(y) \right] \delta(y - z). \quad (2.5)$$

Solution.

$$\mathcal{A}[f(y)]g(y) = -\frac{d^2 g}{dy^2} + fg = \int dz A(y, z)g(z). \quad (2.6)$$

Suppose that $O(x, y)$ is the integral kernel of

$$\mathcal{O} = \left(-\frac{d^2}{dy^2} + f(y) \right)^{-1}. \quad (2.7)$$

b) The operators satisfy $\mathcal{A}\mathcal{O} = 1$. Rewrite this identity entirely in terms of their integral kernels.

Solution.

$$\int dz A(x, z)O(z, y) = \delta(x - y). \quad (2.8)$$

c) Using functional differentiation, prove

$$\frac{\delta F[f]}{\delta f(x)} = O(x, x). \quad (2.9)$$

Hint: Recall the matrix identity

$$\det M = \exp(\text{tr} \log M). \quad (2.10)$$

Solution.

$$\frac{\delta F[f]}{\delta f(x)} = \frac{\delta}{\delta f(x)} \text{tr} \log \mathcal{A} = \frac{\delta}{\delta f(x)} \text{tr} \mathcal{O} \delta \mathcal{A} = \frac{\delta}{\delta f(z)} \int dy dz O(x, y) \delta f(y) \delta(y - z) = O(x, x). \quad (2.11)$$

d) Next show

$$\frac{\delta^2 F[f]}{\delta f(x) \delta f(y)} = -O(x, y)O(y, x). \quad (2.12)$$

Hint: To differentiate an inverse, begin with

$$M^{-1}M = 1. \quad (2.13)$$

Solution.

$$\frac{\delta^2 F[f]}{\delta f(x) \delta f(y)} = \frac{\delta}{\delta f(y)} \text{tr} \mathcal{A}^{-1} = -\text{tr} \mathcal{A}^{-1} \frac{\delta \mathcal{A}}{\delta f(y)} \mathcal{A}^{-1} = -O(x, y)O(y, x). \quad (2.14)$$

e) Finally derive

$$\frac{\delta^3 F[f]}{\delta f(x) \delta f(y) \delta f(z)} \quad (2.15)$$

$$= O(x, y)O(y, z)O(z, x) + O(x, z)O(z, y)O(y, x). \quad (2.16)$$

Solution. Follows trivially from above.

3. Optional: The Gel'fand–Yaglom Formalism

The Gel'fand–Yaglom method can determine functional determinants without explicitly computing every eigenvalue. In favourable cases it is exact, and it can also be implemented numerically. Begin with the Riemann zeta function

$$\zeta(s) \equiv \sum_{k=1}^{\infty} k^{-s}. \quad (3.1)$$

The contour required below is fixed by the argument principle. It is positively oriented and surrounds the positive zeros λ_n of $F(\lambda)$, while avoiding the branch cut of λ^{-s} on the negative real axis.

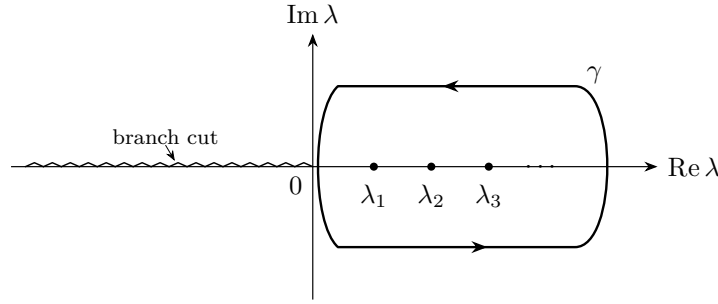


Figure 1. Schematic reconstruction of contour γ in the λ -plane.

Let A have eigenvalues λ_n and eigenfunctions ϕ_n :

$$A\phi_n = \lambda_n\phi_n. \quad (3.2)$$

Associate to it the zeta function

$$\zeta_A(s) \quad (3.3)$$

$$\begin{aligned} &\equiv \text{Tr} \left[\frac{1}{A^s} \right] \\ &= \sum_{k=1}^{\infty} \lambda_k^{-s}. \end{aligned} \quad (3.4)$$

a) Show that zeta regularisation gives

$$\det A = \exp[-\zeta'_A(0)]. \quad (3.5)$$

Because ζ_A and its derivative need not converge near $s = 0$, analytic continuation may be required before this expression is meaningful.

Solution.

$$\det A = \prod_{k=1}^{\infty} \lambda_k, \quad (3.6)$$

while

$$\zeta'_A = \sum_{k=1}^{\infty} -\log \lambda_k \lambda^{-s}. \quad (3.7)$$

Therefore,

$$\det A = \exp[-\zeta'_A(0)]. \quad (3.8)$$

b) Take

$$A = -\frac{d^2}{d\tau^2} \quad (3.9)$$

on $0 \leq \tau \leq 1$, with Dirichlet conditions

$$\phi_n(0) = \phi_n(1) = 0. \quad (3.10)$$

Find the eigenvalues λ_n , then evaluate $\det A$ using

$$\zeta(0) = -\frac{1}{2}, \quad \zeta'(0) = -\frac{1}{2} \log(2\pi). \quad (3.11)$$

Solution. It can be easily found that $\lambda_n = n^2\pi^2$, $n \in \mathbb{Z}^+$. Then,

$$\zeta_A(s) = \sum_{k=1}^{\infty} (n\pi)^{-2s}, \quad (3.12)$$

and

$$\zeta'_A(0) = \sum_{k=1}^{\infty} -2 \log k - 2 \log \pi = 2\zeta'(0) - 2 \log \pi \zeta(0) = -\log 2. \quad (3.13)$$

Hence, we have

$$\det A = 2. \quad (3.14)$$

In cases where the eigenvalues cannot be obtained explicitly, suppose one can construct $F(\lambda)$ whose zeros are precisely the eigenvalues of A .

c) Assume that the $\lambda_n > 0$ are discrete and nondegenerate. Show that

$$\zeta_A(s) = \frac{1}{2\pi i} \int_{\gamma} d\lambda \lambda^{-s} \frac{d \ln F(\lambda)}{d\lambda}, \quad (3.15)$$

where γ is the contour specified in Figure 1 and the branch cut lies on the negative real axis.

Solution. The RHS is given by

$$\begin{aligned} & \frac{1}{2\pi i} \int_{\gamma} d\lambda \lambda^{-s} \frac{d \ln F(\lambda)}{d\lambda} \\ &= \sum_{n=1}^{\infty} \lim_{\lambda \rightarrow \lambda_n} \frac{\lambda - \lambda_n}{\lambda^s} \frac{1}{F(\lambda)} \frac{dF(\lambda)}{d\lambda} \\ &= \sum_{n=1}^{\infty} \lim_{\lambda \rightarrow \lambda_n} (F'(\lambda_n) + F''(\lambda_n)(\lambda - \lambda_n) + \cdots) \left(F'(\lambda_n)(\lambda - \lambda_n) + \frac{1}{2}F''(\lambda_n)(\lambda - \lambda_n)^2 + \cdots \right)^{-1} \\ &= \sum_{n=1}^{\infty} \lim_{\lambda \rightarrow \lambda_n} \frac{1}{\lambda^s} \left(1 + \frac{F''(\lambda_n)}{F'(\lambda_n)}(\lambda - \lambda_n) + \cdots \right) \left(1 - \frac{1}{2} \frac{F''(\lambda_n)}{F'(\lambda_n)}(\lambda - \lambda_n) + \cdots \right) \\ &= \sum_{n=1}^{\infty} \lambda_n^{-s} \\ &= \zeta_A(s). \end{aligned} \quad (3.16)$$

d) Deform γ into the contour γ_- , which hugs the negative real axis as indicated in Figure 2, and derive

$$\zeta_A(s) = \frac{\sin(\pi s)}{\pi} \int_0^{-\infty} d\lambda \lambda^{-s} \frac{d \ln F(\lambda)}{d\lambda}. \quad (3.17)$$

Solution.

$$\begin{aligned}
 \zeta_A(s) &= \frac{1}{2\pi i} \int_{\gamma_-} d\lambda \lambda^{-s} \frac{d \ln F(\lambda)}{d\lambda} \\
 &= \lim_{\epsilon \rightarrow 0^+} \frac{1}{2\pi i} \int_{-\infty}^0 d\lambda (\lambda + i\epsilon)^{-s} \frac{d \ln F(\lambda + i\epsilon)}{d\lambda} - \frac{1}{2\pi i} \int_{-\infty}^0 d\lambda (\lambda - i\epsilon)^{-s} \frac{d \ln F(\lambda - i\epsilon)}{d\lambda} \\
 &= \frac{1}{2\pi i} \int_{-\infty}^0 *0 d\lambda (e^{-s \log \lambda - is\pi} - e^{-s \log \lambda + is\pi}) \frac{d \ln F(\lambda)}{d\lambda} \\
 &= -\frac{\sin \pi s}{\pi} \int_{-\infty}^0 d\lambda \lambda^{-s} \frac{d \ln F(\lambda)}{d\lambda}.
 \end{aligned} \tag{3.18}$$

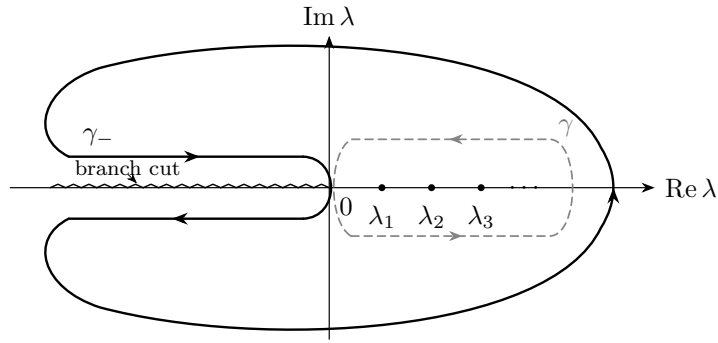


Figure 2. Finite contour deformation.

e) Apply the preceding results to the one-dimensional Schrödinger operator

$$A = -\frac{d^2}{dx^2} + V(x) \tag{3.19}$$

on $0 \leq x \leq 1$, with Dirichlet boundary conditions:

$$\begin{aligned}
 A\phi_n &= \lambda_n \phi_n, \\
 \phi(0) &= 0, \\
 \phi(1) &= 0.
 \end{aligned} \tag{3.20}$$

Assume that the λ_n are positive, discrete, and nondegenerate. Instead of the Dirichlet eigenfunctions, solve the same eigenvalue equation with initial-value conditions

$$\begin{aligned}
 Au_\lambda &= \lambda u_\lambda, \\
 u_\lambda(0) &= 0, \\
 u'_\lambda(0) &= 1.
 \end{aligned} \tag{3.21}$$

Choose

$$F(\lambda) = u_\lambda(1) \tag{3.22}$$

and use the previous parts to establish the Gel'fand-Yaglom theorem

$$\frac{\det A}{\det A_{\text{free}}} = \frac{u_0(1)}{u_{\text{free},0}(1)}, \tag{3.23}$$

where

$$A_{\text{free}} = -\frac{d^2}{dx^2}. \tag{3.24}$$

Solution. From the results above, we have

$$\zeta'_A(0) = \int_0^{-\infty} d\lambda \frac{d \log F(\lambda)}{d\lambda} = \log \frac{F(-\infty)}{F(0)} \implies \det A = \exp[-\zeta'_A(0)] = \frac{F(0)}{F(-\infty)}. \quad (3.25)$$

Note that $F(-\infty) = u_{-\infty}(1)$ is formally infinite. Intuitively, one might guess

$$\lim_{\lambda \rightarrow -\infty} \frac{F(\lambda)}{F_{\text{free}}(\lambda)} = 1, \quad (3.26)$$

which follows

$$\frac{\det A}{\det A_{\text{free}}} = \frac{u_0(1)}{u_{\text{free},0}(1)}. \quad (3.27)$$

The guess is true, as long as $V(x)$ is bounded, when $\lambda \rightarrow -\infty$, $V(x)$ becomes negligible.

f) Use the Gel'fand-Yaglom theorem to compute $\det A$ for

$$A = -\frac{d^2}{dx^2} + \omega^2 \quad (3.28)$$

on the unit interval with Dirichlet boundary conditions. Verify the result independently by finding the eigenvalues and using

$$\sin(\pi z) = \pi z \prod_{n=1}^{\infty} \left(1 - \frac{z^2}{n^2}\right). \quad (3.29)$$

Solution.

$$u_{\text{free},0}(x) = x, \quad u_0(x) = \frac{\sinh \omega x}{\omega}, \quad \det A_{\text{free}} = 2 \implies \det A = 2 \frac{\sinh \omega}{\omega}. \quad (3.30)$$

Meanwhile,

$$\lambda_n = \pi^2 n^2 + \omega^2 \implies \det A = \prod_{n=1}^{\infty} \pi^2 n^2 \left(1 + \frac{\omega^2}{\pi^2 n^2}\right), \quad (3.31)$$

which looks disastrous. However, remember, we can regulate it with

$$\det A_{\text{free}} = \prod_{n=1}^{\infty} \pi^2 n^2. \quad (3.32)$$

Therefore, we find

$$\frac{\det A}{\det A_{\text{free}}} = \frac{\sinh \omega}{\omega}, \quad (3.33)$$

which recovers the Gel'fand-Yaglom theorem result by regulating $\det A_{\text{free}}$ through the analytic continuation of the Riemann function.